

STAT 525

Chapter 20
Two-Factor Studies with One Case per Treatment

Dr. Qifan Song

One Observation Per Cell

- Do not have enough information to estimate **both** the interaction effect and error variance
- With interaction, error degrees of freedom is $ab(n - 1) = 0$
- Common to assume there is no interaction (i.e., pooling)
 - $SSE^* = SSAB + 0$
 - $df_E^* = df_{AB} + 0$
- Can also test for less general type of interaction that requires fewer degrees of freedom

Tukey's Test for Additivity

- Consider special type of interaction
- Assume following model

$$Y_{ij} = \mu + \alpha_i + \beta_j + \theta\alpha_i\beta_j + \varepsilon_{ij}$$

- Uses up only one degree of freedom
- Other variations possible (e.g., $\theta_i\beta_j$)
- Want to test $H_0 : \theta = 0$
- Will use regression after estimating factor effects to test θ , i.e., $Y_{ij} = \mu + \alpha_i + \beta_j + \theta\hat{\alpha}_i\hat{\beta}_j + \varepsilon_{ij}$.
- It is equivalent to model $Y_{ij} = \mu + \alpha'_i + \beta'_j + \theta(\hat{Y}_{ij})^2 + \varepsilon_{ij}$.

Example (Page 882)

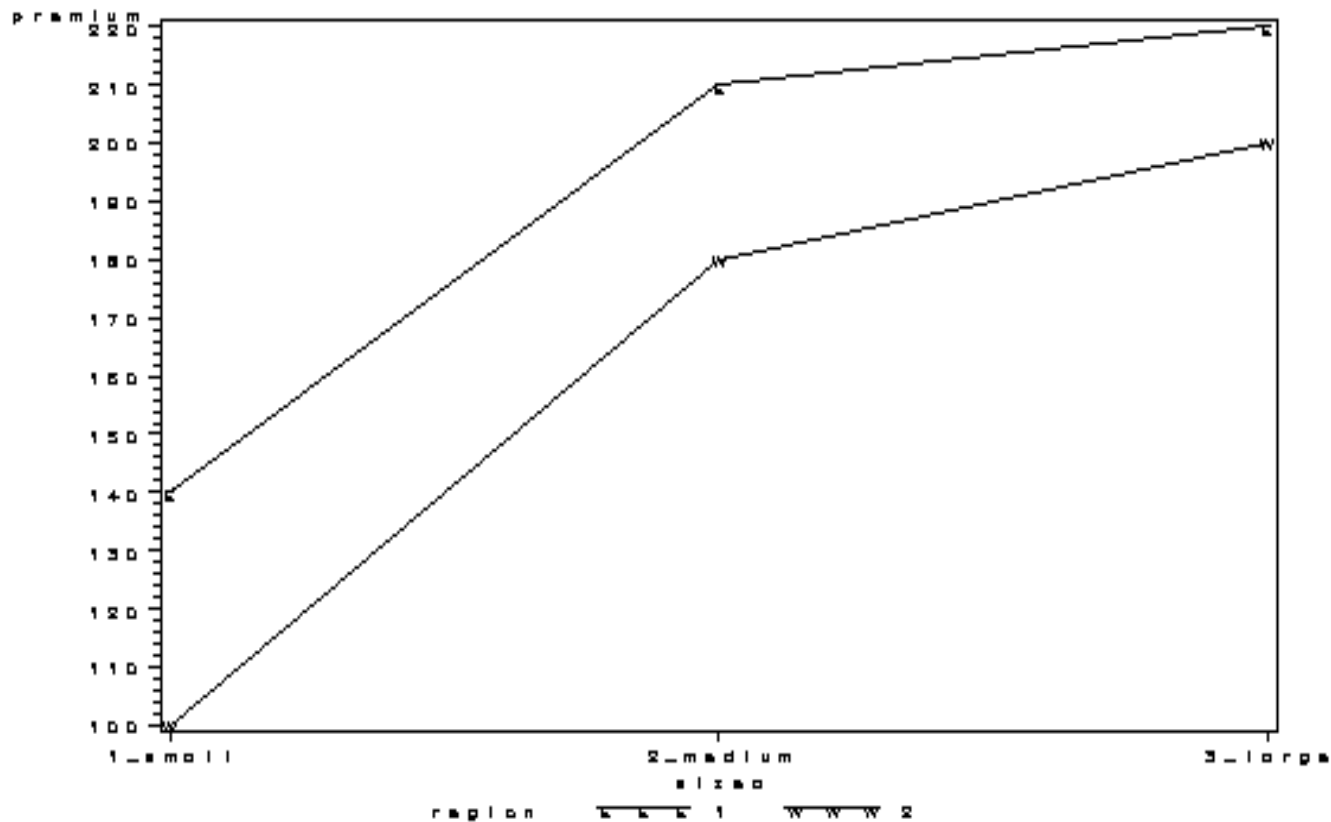
- Y is the premium for auto insurance
- Factor A is the size of the city
 - $a = 3$: small, medium, large
- Factor B is the region
 - $b = 2$: east, west
- Only one city per cell was observed

```

data a1; infile 'u:\.www\datasets525\CH20TA02.txt';
  input premium size region;
  if size=1 then sizea='1_small ';
  if size=2 then sizea='2_medium';
  if size=3 then sizea='3_large ';

symbol1 v='E' i=join c=black; symbol2 v='W' i=join c=black;
proc gplot data=a1;
  plot premium*sizea=region/frame;
run; quit;

```



```
proc glm data=a1;
    class size region;
    model premium=size region ;
output out=diag P=yhat;
run;
```

```
proc glm data=diag;
    class size region;
    model premium=size region yhat*yhat;
run;
```

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	10737.09677	2684.27419	208.03	0.0519
Error	1	12.90323	12.90323		
Corrected Total	5	10750.00000			

R-Square	Coeff Var	Root MSE	premium Mean
0.998800	2.052632	3.592106	175.0000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
size	2	9300.000000	4650.000000	360.37	0.0372
region	1	1350.000000	1350.000000	104.62	0.0620
yhat*yhat	1	87.096774	87.096774	6.75	0.2339

We fail to reject the additive hypothesis.

One Quantitative Factor

- Similar to regression with one indicator or categorical variable
- Plot the means vs the quantitative factor for each level of the categorical factor
- Based on this plot,
 - Consider linear/quadratic relationships for the quantitative factor
 - Consider different slopes for the different levels of the categorical factor
 - Can perform lack of fit analysis
- If two quantitative variables, can consider linear and quadratic terms. Interactions modeled as the direct product. Lack of fit test very useful. Again very similar to linear regression models.

Chapter Review

- Two-Factor Studies with $n_{ij} = 1$
 - No degrees of freedom for interaction
 - Tukey's test for additivity
 - * Use only one degree of freedom
 - * Can be generalized to use more degrees of freedom
- One or both factors are quantitative
 - A test for interactions effects can be obtained by regression methods
 - * Include interactions by taking direct products